

Overview:

Memo	Date	Description	Date Due
<u>ProjLab4</u> - Design a Better Product • Apply human factors data and engineering design to redesign your hand tool. • Do not change the user that you have (e.g. the typical fixit person to a construction worker).	30/31 Oct. 2025	1. Create a single “best” design. This can be developed using a CAD-type tool, 3D printing or by hand (but should be professional). If you are doing a redesign, include both before and after drawings or diagrams. If you are developing a new design, please include some drawing, etc. representation of the product that are currently being used that inspired your design. 2. <u>Document and provide HFE rational</u> for your new design. 3. Include <u>specifications</u> so that your redesign can be given to a manufacturing engineer or mechanical engineer/designer and used directly. Note specifications also include the materials needed. 4. Delineate any HFE design tradeoffs in your analysis and redesign (a table may be helpful in presenting this information).	28 Nov. 2025

Resources**1. Relevant Chapters:**

Below are some relevant chapters that may help in the redesign of your hand tool. You may want to skim back through the chapters to find things that may be relevant to your tool.

- Week 2 Lecture, Chapter 2 of Textbook: Research Methods Though it will not be feasible to test and evaluate the design, provide a suggested experimental design to evaluate the goodness of the design.
- Week 2 Lecture, Chapter 3: System Design and Task Analysis -consider displays, labels, and instructions (on product)
- Week 3 Lecture, Chapter 10: Anthropometry
- Week 12 Lecture, Chapter 8: Visual Displays
- Week 12 Lecture, Chapter 9: Control and Compatibility

2. Use your last three project labs, in particular, your Task Analysis, Market Analysis, Survey analysis and Literature Review. Start with the labs associated with this product. Now that you’ve had a chance to learn a little more about products that people use, i.e., anthropometry, controls and displays, think through the following:

a. Reconsider the users for whom you are designing. Do you need to change, redefine, or expand the users of the product? How might this affect your market analysis? Remember that the market analysis should consider competitors, price differences, feature differences, specification differences, target users, and tradeoffs.

The primary users of the redesigned silicone pasta strainer are still individuals cooking in the kitchen, particularly adults who regularly cook. However, because the new design incorporates improved safety, grip, and stability, the potential user group can be expanded to include younger or less experienced cooks, older adults, and individuals with limited hand strength. This shift affects the market analysis by broadening the target audience and positioning the product as more inclusive and user-friendly. Since the redesign addresses common pain points such as pasta slipping out and the risk of burns, the product becomes more marketable and gains a stronger competitive advantage over other strainers that do not resolve these issues. At the same time, the addition of new features may increase production costs, which could raise the retail price and create tradeoffs between adorability and performance. Considering competitors, pricing, features, specifications, target users, and tradeoffs helps show how expanding the user group makes the product more competitive while still acknowledging the potential impact on cost.

b. Consider also the functionality of your product. Should it be expanded, reduced? Remember one of the things to consider when designing a product is how people may use that product for purposes that it is not intended.

The core function of our product is to let users safely drain boiling water directly from a pot while minimizing the risk of burns and spills. The current design already supports this primary function through these three elements: 1. A curved clip-on strainer sized to fit common pot diameters. 2. two clip styles for single-handle and double-handle pots 3. Heat-resistant handle covers add-on and removable, to improve grip and thermal protection.

Given our HFD goals: safety, simplicity, low cognitive load, we don't want to over-expand the functionality and turn it into a multi-tool. Adding too many features would increase weight, complexity, and cognitive load/error potential, which conflicts with what we learned from task analysis.

However, small and controlled expansion of functionality is reasonable, for example: 1. Not only strain pasta, but more kinds of hot food (This is why we adjust hole sizes and intervals, to not only strain food in a quick way, but also not let food out or stick on the holes). 2. The handle covers may be used for other pot or pan handles without strainer.

Anticipated unintended uses: 1. Using the strainer as a full lid or splatter guard when cooking for a long time. 2. Using it in the sink as a standalone colander, like washing vegetables or fruits. 3. Using handle covers as standalone heat-resistant mitts for hot and heavy cookware like an oven plate. 4. Put the product in an oven broiler or open flame.

From HFD perspective, these unintended uses could raise safety issues (excessive heat exposure, structural failure or deformation). As a result, we will explicitly limit functionality in our specifications and labelling things including specify maximum pot size, load, and temperature for safe use; and add simple icons/text such as “draining only, not a full lid” “do not put on open flame or in an oven.”

c. Given your expanded knowledge of products that people use, i.e., anthropometry, controls and displays, revisit the potential risks and problems and see how you might modify these when redesigning the product. Are there some other literature sources you may need to identify?

- Clip fit may fail on pots outside common diameters; redesign using 5th–95th percentile hand sizes and measured pot rim radii to ensure stable attachment.
- Handle covers may not fit all hand spans or grips; size grips using Bodyspace/CAESAR anthropometric data to support secure handling with both small and large hands.
- Users may misinterpret product as a lid or oven-safe tool; add simple icons (“draining only,” “no oven/flame”) plus a tactile click to indicate correct attachment.
- Steam path may expose the user’s hand; add a 15–25 mm steam-deflector lip to increase distance between steam and fingers.
- Thermal deformation possible with plastic clips; specify high-temp polymers (e.g., glass-filled nylon) and label max safe temperature (e.g., “≤230°C”).
- Users may attempt to strain foods too small for hole geometry; include two snap-in plates (coarse for pasta, fine for rice/veg) while keeping mechanism simple.

Additional Literature;

- **CAESAR anthropometric database / Bodyspace textbook** for designing hand grips and reach envelopes.
- **Thermal injury threshold literature** (e.g., time-temperature burn studies) to determine safe handle temperature limits.
- **Cookware safety regulations (FDA, CPSC, Health Canada)** to justify material choice and temperature ratings.

Dive in and Redesign

1. Apply human factors data and engineering considerations. Note that there are certain human factors engineering principles and knowledge to employ, but this is also about design so there is latitude for creativity associated with coming up with a better design;

however, it must be backed with human factors principles and data. Further that it must be fully justified.

For our final product, we created a strainer that is essentially “error-proof” for our customers. We did this by adding the following design considerations:

1. **A steam guard-** The steam guard will prevent users from burning their hands by blocking the steam from the user’s hands. They are placed where the handles would be, that way the user will not have any accidents.
2. **Rubber Handles-** The handles will further prevent burns by covering metal handles. This will help make sure that customers do not burn their hands by touching the hot metal.
3. **Strainer clips-** While the strainer is meant to simply slip onto the pot, we added clips to the side to prevent the strainer from falling off. That way, hot water will not splash onto the user, further preventing injuries.

2. Remember we are looking for specifications that will match your target customer. The specifications should be such that the design can be given to mechanical/electrical engineers for final design.

The product will be made from silicon rubber. They will be manufactured using liquid injection molding. The dimensions of the product, which are listed in the drawing, will be manufactured to match the dimensions of a typically sized pot.

3. You will want to include a drawing(s) and/or diagram(s) of your redesign. These should include the necessary dimensions. You may choose to demonstrate how the tool fits in the hand. Display features of the redesign.



